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Q1 Docker Setup Guide

Getting Started

A video tutorial has been created to help walk you through the steps in this document. The setup video if you have:

- Mac: https://youtu.be/fHej6YzEupE
- Windows 10: https://youtu.be/p31bIzgPsIM

To get started with Docker follow the documentation below.

1. The first step to get started is to download and install Docker according to the instructions for your operating system found here: https://docs.docker.com/get-docker/
   (a) Windows 10/11: https://docs.docker.com/docker-for-windows/install-windows-home/
      i. Before installing Docker on Windows 10/11 Home, be sure the system requirements are met. Specifically, ensure Windows is updated to version 2004 or higher and WSL 2 is enabled by following instructions from Windows Subsystem for Linux Installation Guide for Windows 10.
   (b) Mac: https://docs.docker.com/docker-for-mac/install/
   (c) Linux: https://docs.docker.com/engine/install/
      i. GUI: https://docs.docker.com/desktop/install/linux-install/
2. After running Docker, open a terminal (Windows users **use Powershell** not CMD) inside your HW3 directory.
3. From inside the HW3 directory run the following command for MacOS:

```
docker container run -d \
-v "${PWD}/Q1:/root" \
-p 127.0.0.1:6242:8888 \
--name hw3 \
polodataclub/cse6242hw3-public:Q1
```

   For Windows, run the following command:

```
docker container run -d `
-v "${PWD}/Q1:/root" `
-p 127.0.0.1:6242:8888 `
--name hw3 `
polodataclub/cse6242hw3-public:Q1
```

   or alternatively as a single line command.

(a) Note: Linux and Mac users may require adding sudo before docker commands.

(b) Macbooks: You may this warning "WARNING: The requested image's platform (linux/amd64) does not match the detected host platform (linux/arm64/v8) and no specific platform was requested", This is okay and should not affect functionality for this question.

If you're encountering errors for MacOS when running cell 2 for "Initialize Pyspark Context" please follow the steps below:

i. Go to Docker Desktop > Settings > General
ii. Uncheck :Use Rosetta for x86_64/amd64 emulation on Apple Silicon".
iii. Click Apply & Restart.
iv. Rerun the docker container using the previous command from Step 3

(c) The first time you run this command it will download the polodataclub/cse6242hw3-public image from DockerHub: https://hub.docker.com/r/polodataclub/cse6242hw3-public. The Docker image is approximately 1GB so the time it takes to download will depend on your internet connection.

(d) The command is reproduced below on the left with line breaks and shorthand options extended, plus comments to the right explaining what each part does:

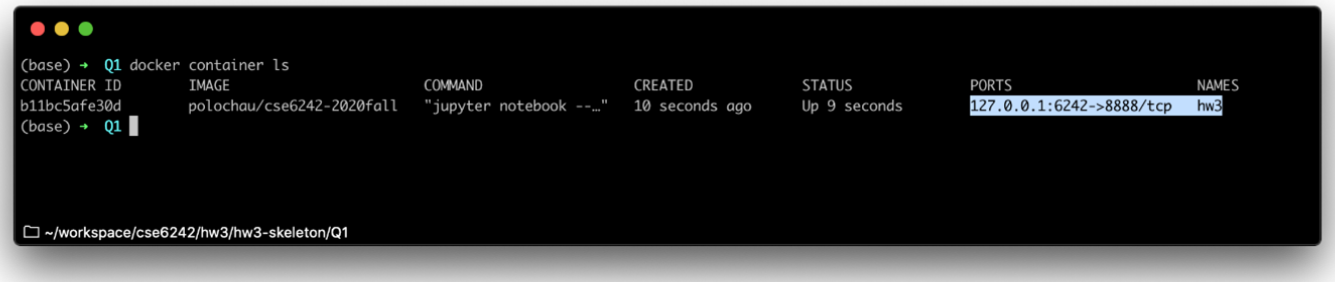
`docker container run \`
`  --detach \`
`  --volume ${PWD}/Q1:/root \`
`  --publish 127.0.0.1:6242:8888 \`
`  --name hw3 \`
`  polodataclub/cse6242hw3-public:Q1`
	`# Start running a container`
	`# Run in the background`
	`# Make Q1 dir available at /root`
	`# Forward port 6242 to 8888`
	`# Name the container "hw3"`
	`# Use this container image`

(e) This command will start a container with the name hw3. You can change the name to whatever you'd like, but the remainder of the commands in this document assume the container name is hw3

(f) The -p flag will publish port 8888 on the container to port 6242 on the host. This will allow you to access Jupyter by loading http://localhost:6242 in your browser, which is set to accept connections on port 8888 inside the container.

(g) The -v flag will share a local directory on your computer (which should be the HW3 directory's Q1 subdirectory) with the /root directory on the container, which is the directory that Jupyter will start in, making your files accessible.

4. At this point you can run docker container ls to confirm the container is started.



5. Once the container has been started you may go to http://localhost:6242 in your browser to access Jupyter. Confirm that the csv file is uploaded to same folder as Q1.

Helpful Commands

- For Windows users that prefer CMD prompt, the example command to create and start the container would be:

```
docker container run -d ^
-v "%CD%/Q1:/root" ^
-p 127.0.0.1:6242:8888 ^
--name hw3 ^
polodataclub/cse6242hw3-public:Q1
```

- Show all running containers:
  - docker container ls
- Show all containers (started and stopped):
  - docker container ls -a
- Force kill and remove a container (useful if you made a mistake and need to start a new container with the same name):
  - docker container rm -f <container>
  - Ex: docker container rm -f hw3
- Show all images:
  - docker image ls
- Attach to a running container (type exit to detach from the container):
  - docker container exec -it <container> /bin/bash
  - Ex: docker container exec -it hw3 /bin/bash
- Start a stopped container:
  - docker container start <container>
  - Ex: docker container start hw3

FAQs

1. After I restart my computer, I am unable to attach to the container
   (a) After restarting your computer the Docker container may not automatically restart. See the helpful commands above on how to start a stopped container. Additionally, check to make sure the Docker daemon/service is running on your host platform. If it doesn't start automatically, find and run the Docker desktop application, which should start the Docker service.
2. How do I transfer my code to the docker container?
   (a) There is no need to manually transfer code between your local environment and the docker container. Since the container was created with a bind mount using the -v flag, all the files are automatically synchronized between your local environment and the container. If you need to access a file inside the container, make sure you place it at the point that you bound the volume (The Q1 subdirectory, if using the example command in the instructions above)

3. I'm getting an "Error response from daemon: Conflict. The container name "/hw3" is already in use by container" error. How do I fix it?

   (a) This error typically means you already have a container named "hw3." Instead of creating a new container, you should list all containers and either start the container if it's stopped (status is Exited) or attach to it if it's running (status is Up). If it is misconfigured and you need to completely remove the "hw3" container and create a new one, you can do so by killing the container and removing it using the commands above

4. I'm getting an "Error response from daemon: driver failed programming external connectivity on endpoint" error. How do I fix it?

   (a) This typically means you have another application (or another container) using the port specified in the -p flag. Check that you don't already have a hw3 container running, or try changing the port from 6242 to 8888 and use that to access Jupyter instead.

5. While installing Docker, I get an error "Installation failed: one prerequisite is not fulfilled"

   (a) For Windows 10 Home, Docker is only compatible with Build 19018+. So you can check from System Information before installing Docker and update Windows if required.

6. When installing WSL 2 on Windows, I'm getting the error "Elevated permissions are required to run DISM."

   (a) When installing WSL 2, you will need to do so as an administrator. You can start an elevated command prompt by accessing the Power Users menu by pressing Windows+X or right-clicking the Start button. On the Power Users menu, choose "Command Prompt (Admin)" and run the WSL 2 installation commands.

7. When trying to upgrade WSL, I'm getting the error: "This update only applies to machines with the Windows Subsystem for Linux."

   (a) There are a few reasons you will see this error. The most common reason is that "after you enabled WSL, a reboot is required for it to take effect, reboot your machine and try again." Additional details may be found in the Windows Subsystem for Linux Installation Guide for Windows 10

Additional Notes:

1. Docker containers should be treated as ephemeral. In situations where the Docker bind mount does not correctly work and the Jupyter notebook is not shared between the Docker container and the local machine, the Jupyter notebook may be uploaded manually as a workaround. However, since the containers should be treated as ephemeral, you must manually download the notebook by navigating to File -> Download as -> Notebook otherwise the notebook may be deleted and work may be lost.
2. There have been issues when running Docker inside a directory that is synced with OneDrive or Dropbox.

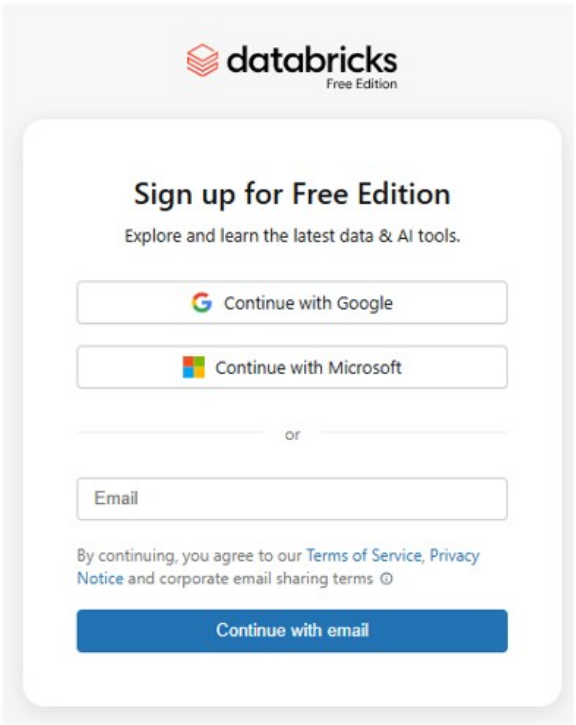
Q2 Databricks Setup Guide

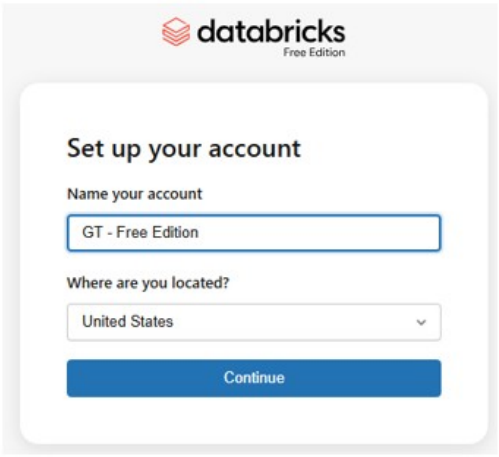
Getting Started

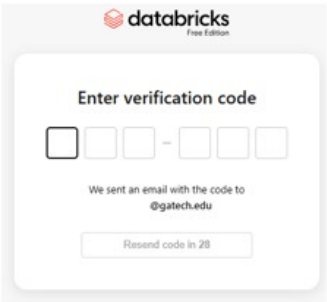
For Q2, we will use the Databricks platform to execute Spark tasks. Databricks has excellent documentation on how to get started - please refer to this for more details on the steps listed below and to assist with troubleshooting. Follow these steps to get started:

1. Go to this website to create a Free Edition account on Databricks.
2. Select "Sign up for Free Edition" to create an account. Enter your email - please use your school email (***.gatech.edu) email to sign up for this platform. Select "Continue with email".

   Note: Do NOT select the Free Trial option - this may cause errors and has credit limitations.
3. Check your email for verification code and follow the prompts to complete the setup process.







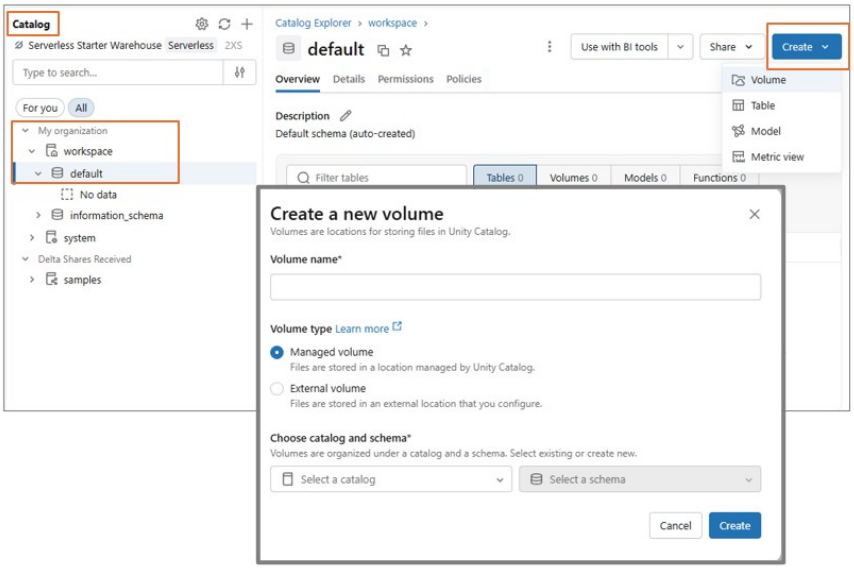
4. After account set up, use this website to login your account.

Note: As a Free Edition user, your access is credit based and the credit is renewed daily (Databricks has assured us that it's highly unlikely users will use up all credits in a 24-hour period, however if this does happen, the credits will be restored the following day.

Create Volume & Import Data

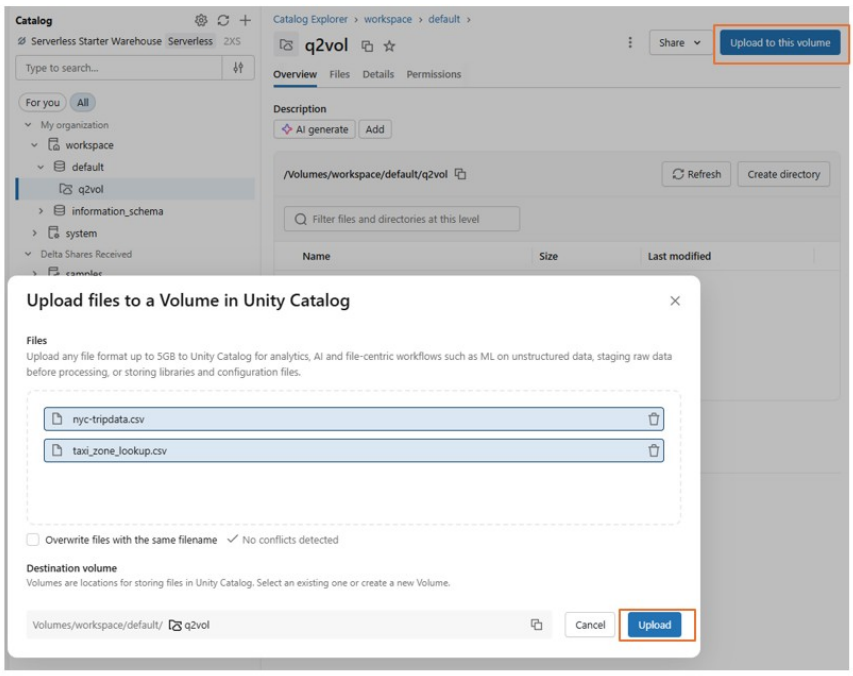
Databricks uses "Volumes" to manage access and storage of data. To import data, we must first create a volume - for this exercise, we recommend naming the **q2vol** (since this will be referenced in the provided **q2.dbc**)

1. On the left side bar, in Databricks, select **Catalog** → **My organization** → **Workspace** → **Default**. A blue create button should show on the right, select **Create** to create a volume. Enter volume name q2vol, select **Managed volume**, Under Choose catalog and schema - select **workspace** and **default**, then **create**.



Note: It may take a few minutes for the volume to be created, you may need to wait and/or refresh the page before moving to the next step.

2. Next import the Q2 data files into the volume, select **Upload to this Volume**, then follow the prompt to drop or browse for the data files, then select **Upload**.

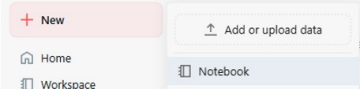


Import Notebook and Attach Cluster

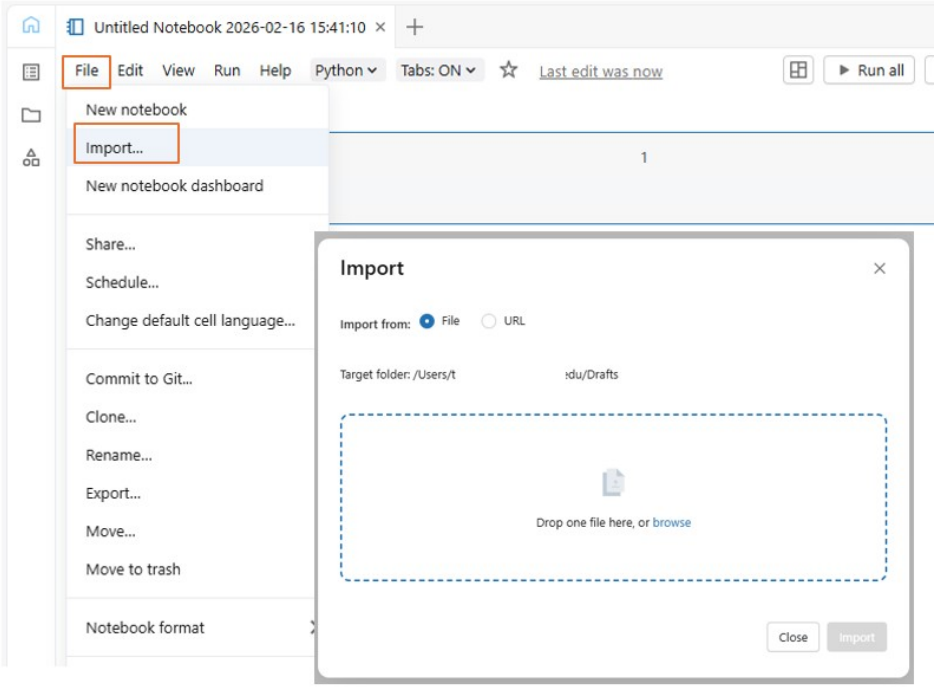
Databricks Free Edition is a serverless environment, to run notebooks you will need to create and/or import your notebook into the Databricks environment

1. To import the q2.dbc notebook (this is a template notebook containing the starter code for Q2 in **hw3-skeleton/q2**).

   (a) Select **New**→ **Notebook** to create an untitled blank notebook.

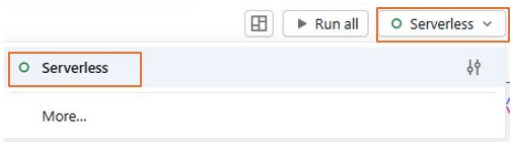


   (b) From within that new notebook, go to **File**, select **Import**. Import the template notebook, **q2.dbc** into your workspace.



   (c) To access imported files, select **Workspace**, you should see a list of available files, select the **q2** file to open the notebook.

2. At the top of the notebook, select **Serverless** (twice) to attach the imported notebook to a cluster. Refer to this document for more information: Customize Notebook Appearance

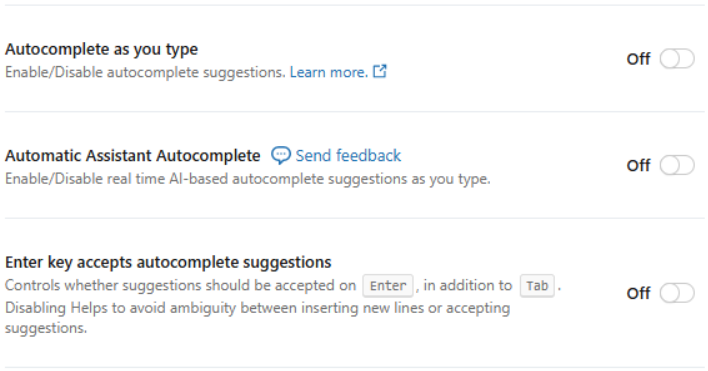


3. Within your notebook, you can access the uploaded data file **nyc-tripdata.csv** and **taxi_zone_lookup.csv** using the Databricks file system utilities. The code snippet to read the data is already provided in the dbc file present in the hw-3 skeleton.

Turn off Built-in AI Assistant

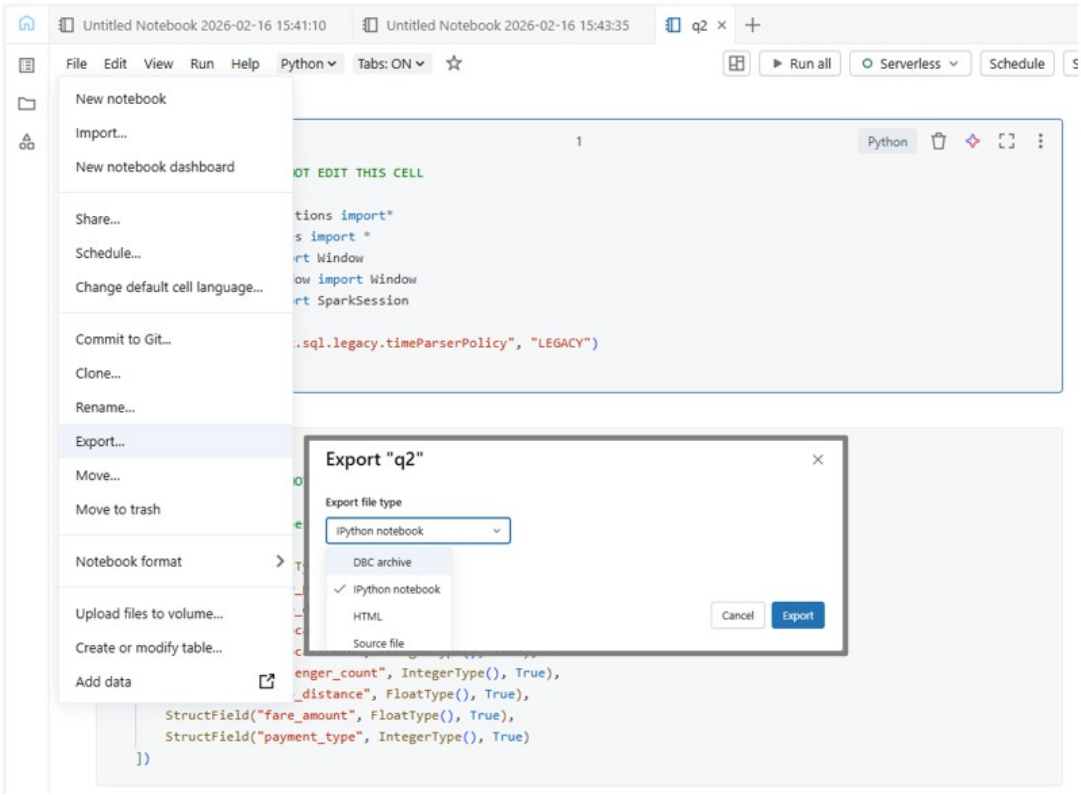
To ensure your work is in compliance with course AI policies, please follow the steps below to turn off Databricks built-in AI Assistance.

- Click your username/icon (upper-right corner of workspace)
- Select **Settings** → **Developer**
- Under the **Code Editor** Toggle **OFF** Autocomplete and any other listed AI/Assistant Settings



Exporting Files

To export the q2.dbc and q2.ipynb deliverables, select **File** → **Export** → **DBC archive** or **iPython Notebook** (ensure these files are correctly named per the homework instructions).



Setup Guide [For Q3]

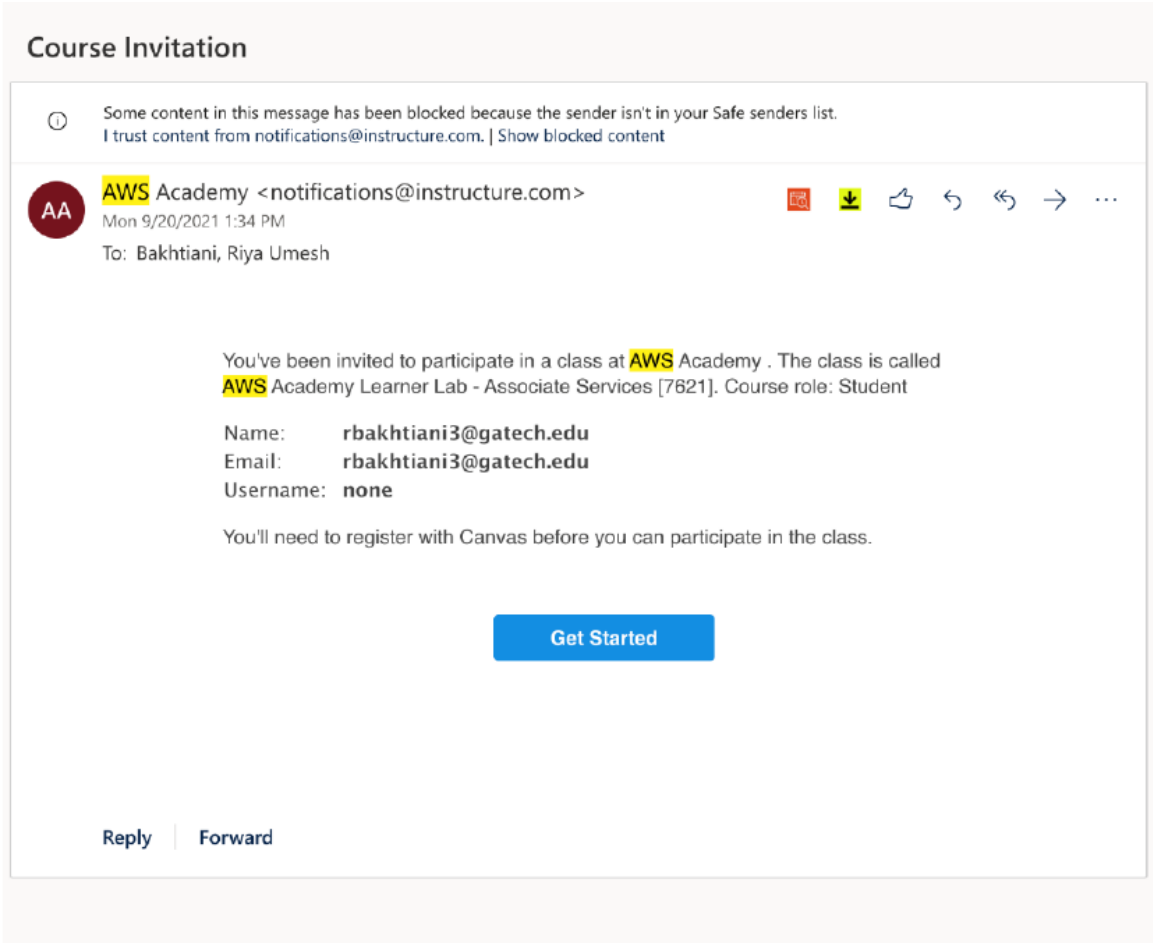
Getting Started

Q3 AWS Setup Guide

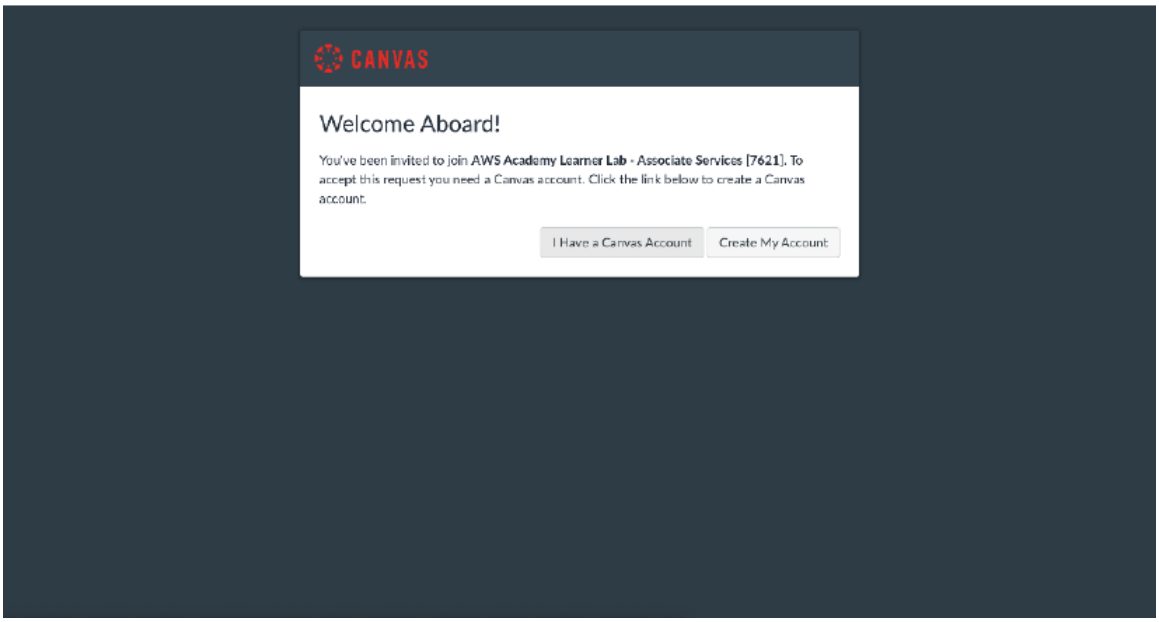
1. Create an AWS Academy account

You will receive an email from notifications@instructure.com inviting you to participate in the **AWS Academy Learner Lab – Associate Services** course. Your AWS Academy allows you to access EC2, Elastic MapReduce and S3 storage. Click on the button to **Join AWS Academy** in the email to proceed.

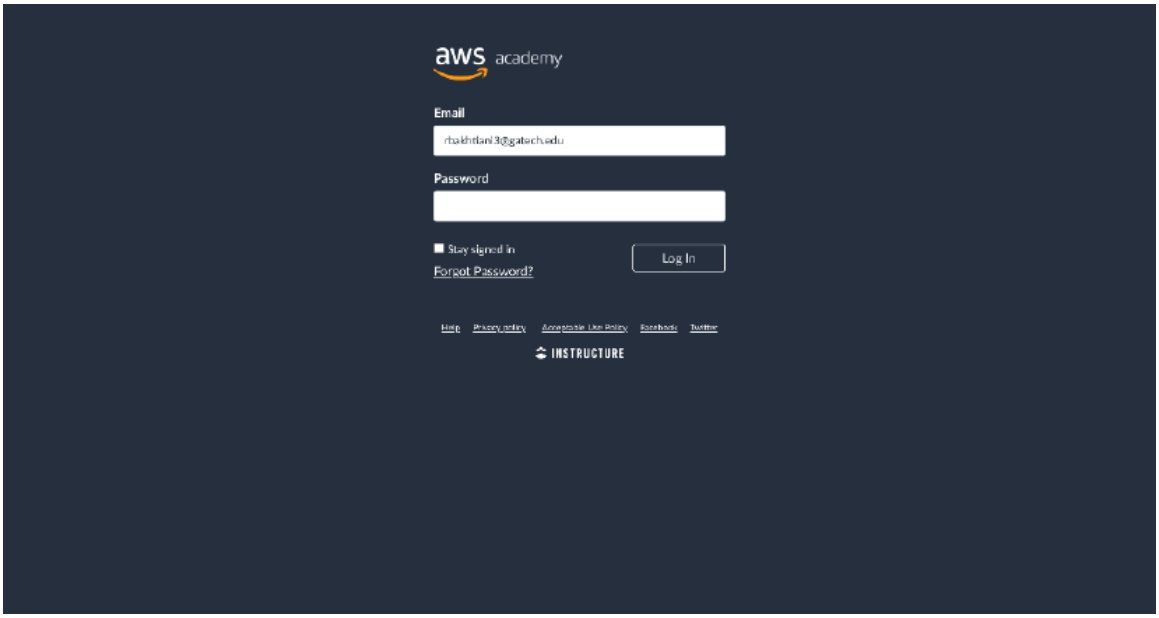
Note: AWS Academy does **NOT** work on Safari. Use a Chrome-based browser for the best results.



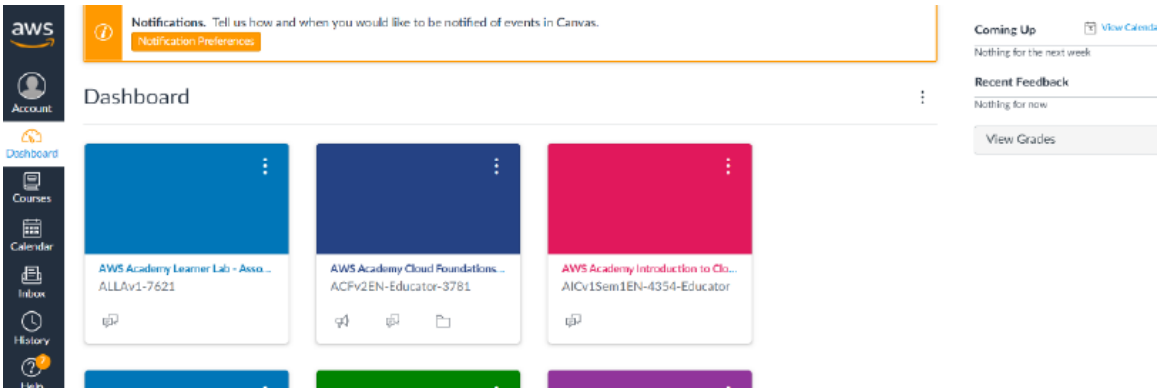
You will be taken to the Canvas page. When it appears, click on the "Create My Account" button.



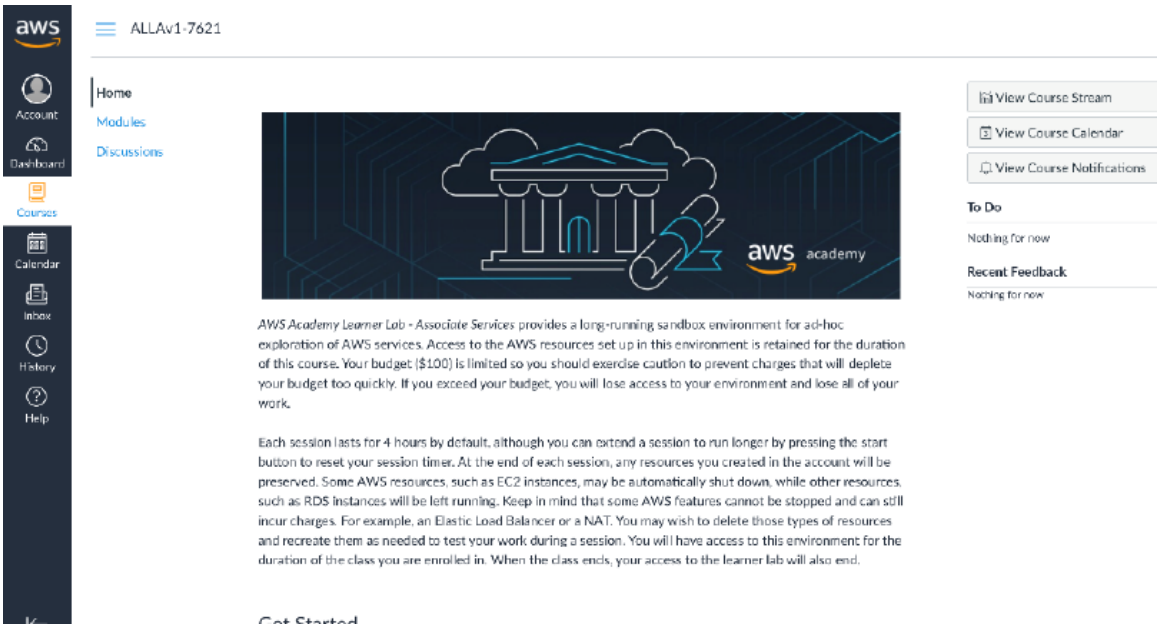
Now, fill in the requested information (e.g. e-mail address, password, etc.). Once submitting, you'll be able to log in to your account at the following URL: `https://awsacademy.instructure.com/login/canvas`
When you arrive at the login URL you will see a screen like this:



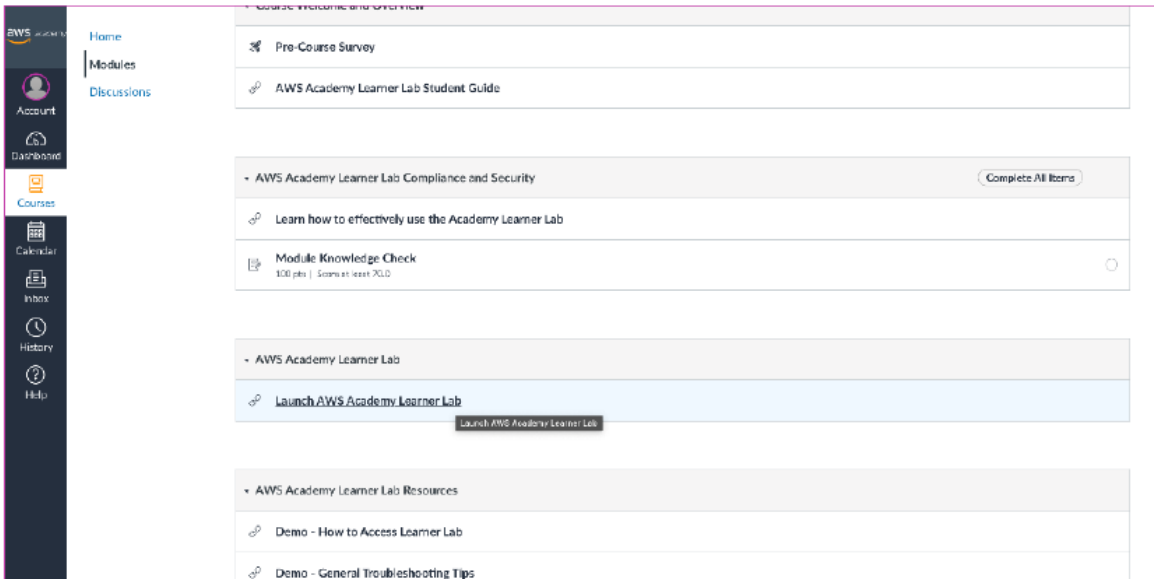
Please enter your e-mail address and password used when setting up the AWS Canvas account (not the Canvas account for the CSE 6242 course). Click the "Log In" button. You will then see the AWS Canvas home screen. Click on the "Dashboard" tab on the far left. You should see a screen like this one (although the number of courses on your screen will differ):



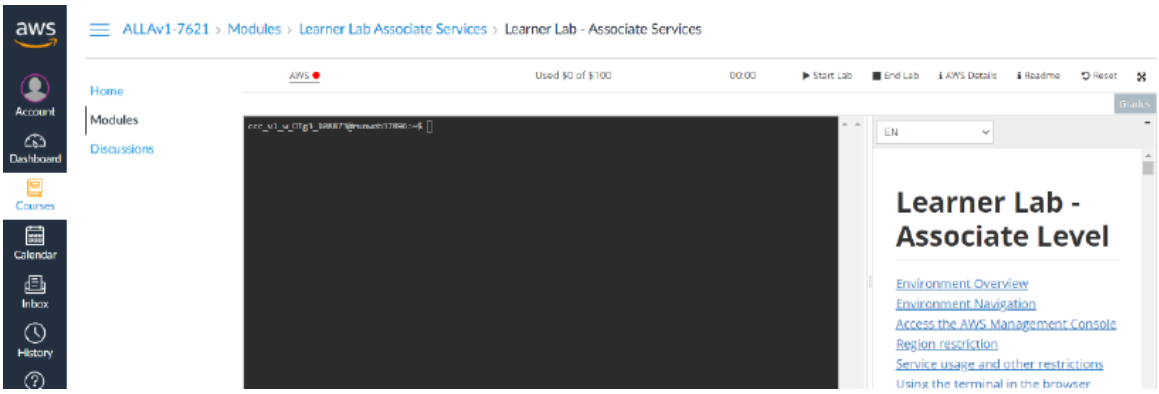
Click on the "AWS Academy Learner Lab" course button. You should see the following screen (the course homepage).



Click on the "Modules" link on the near-left menu. A list of course modules will appear:

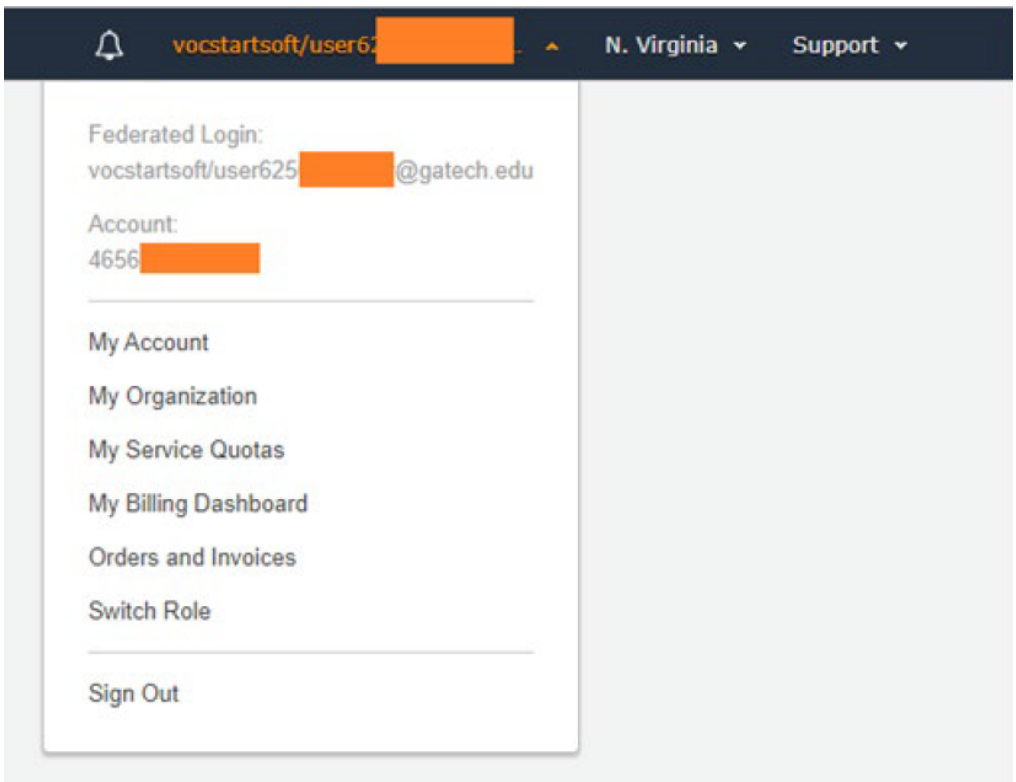


Click on the "Launch AWS Academy Learning Lab". You'll see a screen like the following screen:



Next, click on the "Start Lab" button on the top right. You'll see an agreement from Vocareum. Scroll to the bottom of the agreement and click "I agree." Next, click "Start Lab" at the top. It will take a few minutes for the account to be set up. Once the red circle next to AWS turns green in top left, click the Green button and a new window will open.

You are now in the AWS management console! It will look something like this [right top corner].



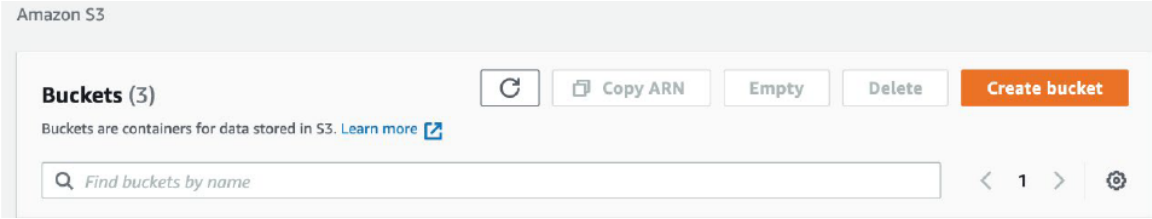
If you have any problems with this process, please let the course staff know on EdStem via the dedicated AWS Setup thread.

2. Create storage buckets on S3

We need S3 for storing the final results of the assignment which will be submitted to the autograder. Data (or objects) in S3 are stored in what we call "buckets". You can think of buckets as folders. All S3 buckets need to have unique names.

1. In the AWS Management Console click on **S3** under **All services** → **Storage**.

   In the S3 console, click on **Create Bucket**.



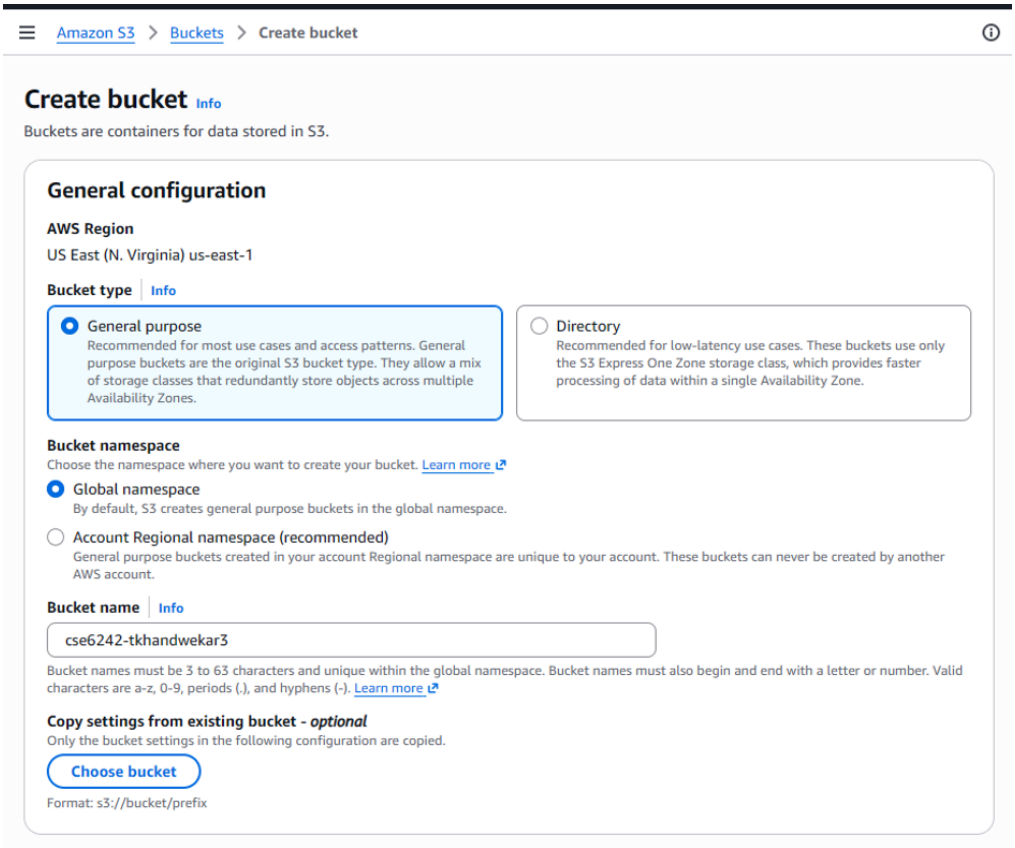
2. Create a bucket: Enter the following details (bucket name and region) then click **Create Bucket** at the bottom of the screen. **Keep all other settings the same.**

   Bucket Name Format: cse6242-<GT username>

   Example: cse6242-gburdell3

   Region: US East (N. Virginia)

   VERY IMPORTANT: Please select "US East (N. Virginia)" only. If you have buckets in other regions, data transfer charges would apply.



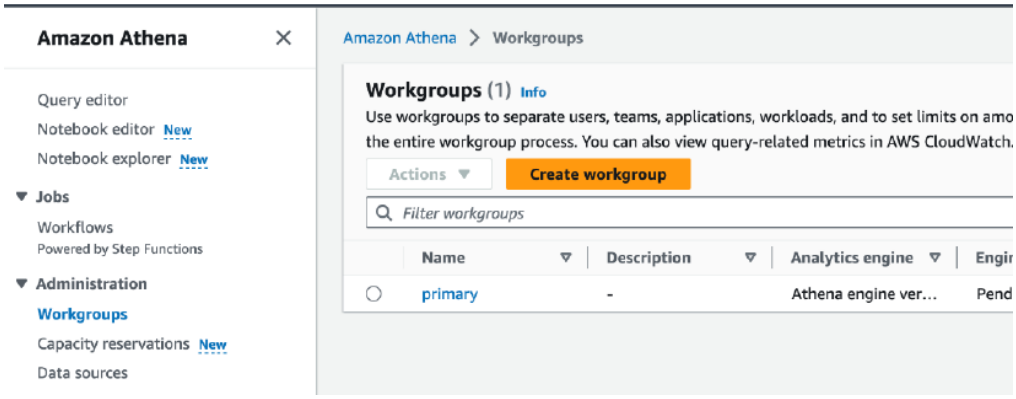
3. A new bucket will appear in the S3 console. Clicking on it will show you that it is empty.

We are done with creating S3 buckets at this point.

3. Creating an Athena Workgroup

This section will cover creating a Workgroup in Amazon Athena. For further information about workgroups in Athena, click here.

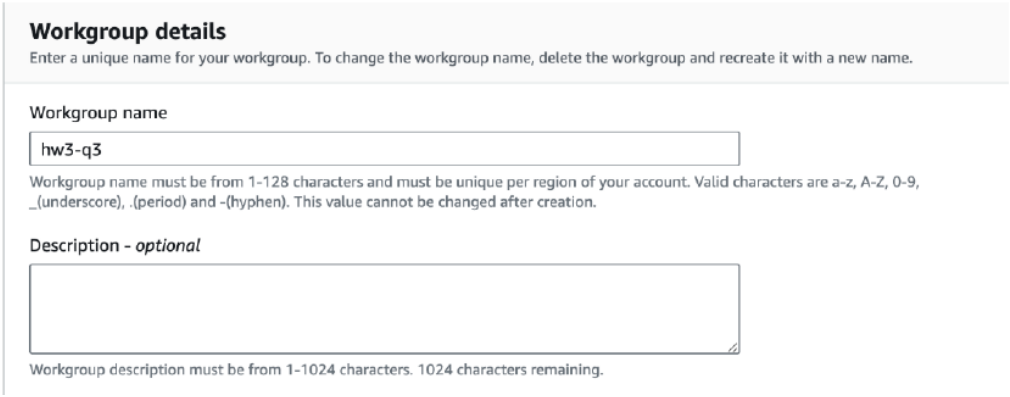
1. Go to Amazon Athena. Select Workgroups on the left menu. Click "Create workgroup".



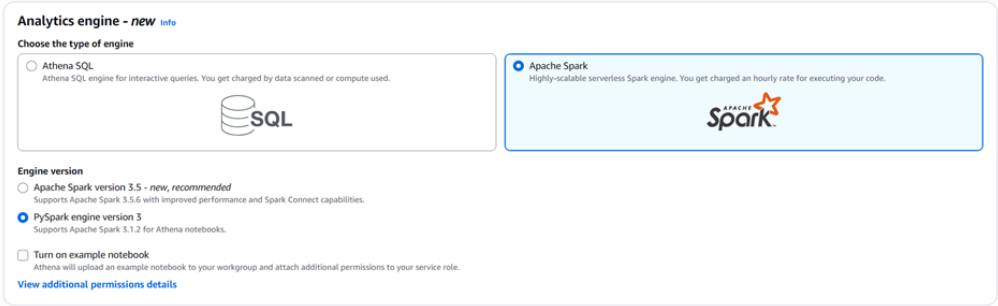
2. Make sure the region specified in the top-right corner of the page is **N. Virginia**. Otherwise click on it and from the drop-down choose N. Virginia.

3. We will now fill out the various configuration fields to create a new Workgroup:

   a. Give your workgroup a name. It can be anything you want.

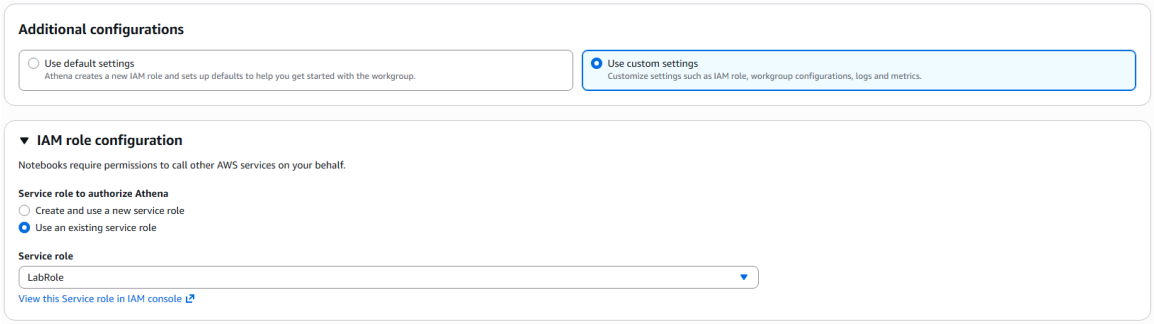


   b. Select “Apache Spark” as the Analytics Engine



Important: As of now, Athena defaults to Spark Version 3.5, which **does NOT** support notebooks. You must explicitly select **Spark Version 3.0** to proceed.

c. Under Additional Configurations, select “Use custom settings” and then choose “LabRole” as the existing IAM service role.

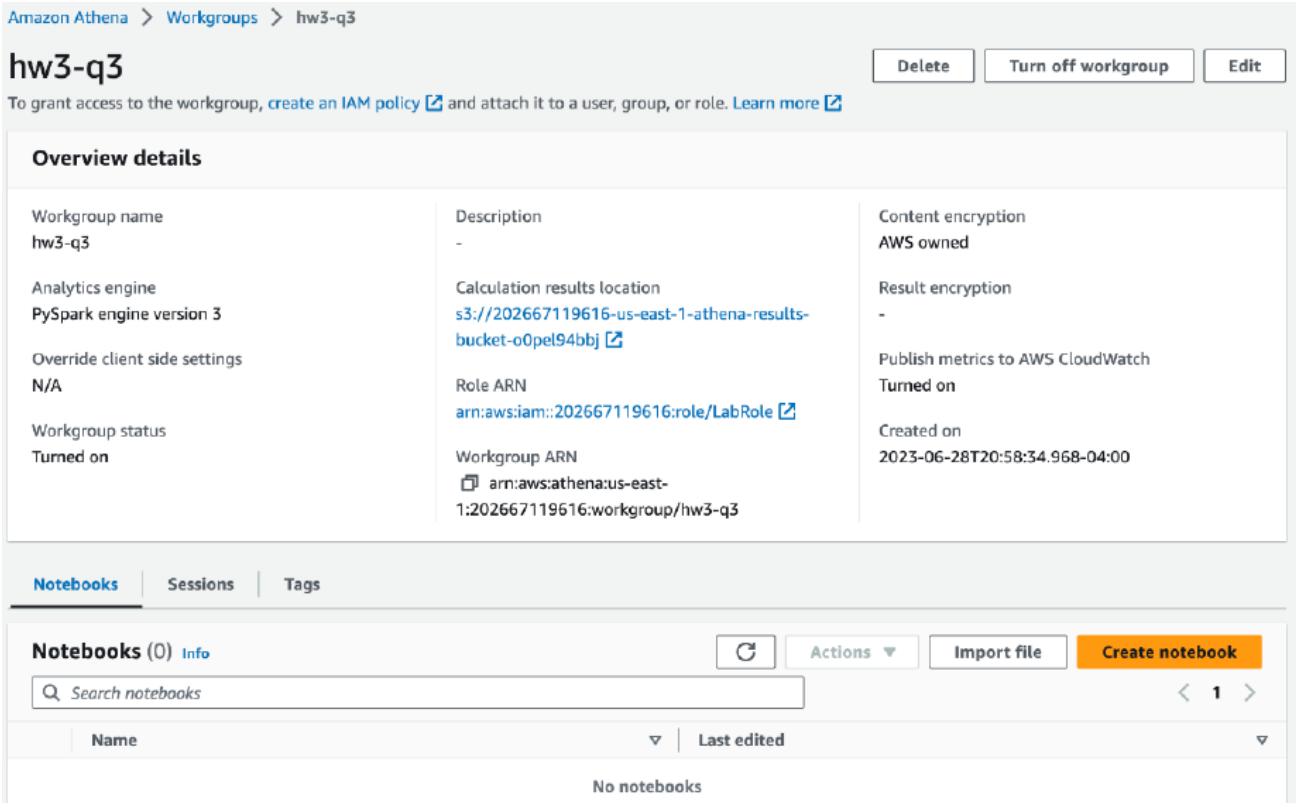


d. Import the hw3-q3 skeleton notebook.

4. Get started with the skeleton

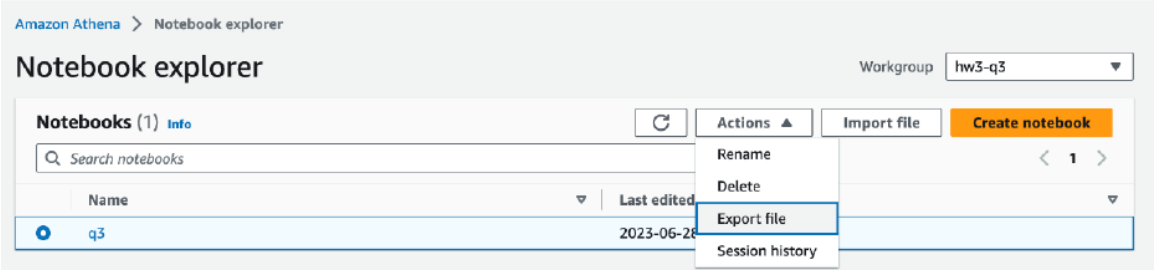
In this section we will import the skeleton file to the workgroup and run our first cell.

1. Select your newly created workgroup from the list and then click "Import file". Import the q3.ipynb file provided in the skeleton.



2. Run the first code cell which will import PySpark, followed by a cell which should contain sc to start the Spark Application so you can start programming the assignment.

3. Once you have finished coding, you can export the notebook from "Notebook Explorer" by selecting your notebook and clicking "Export File" from the Actions dropdown.



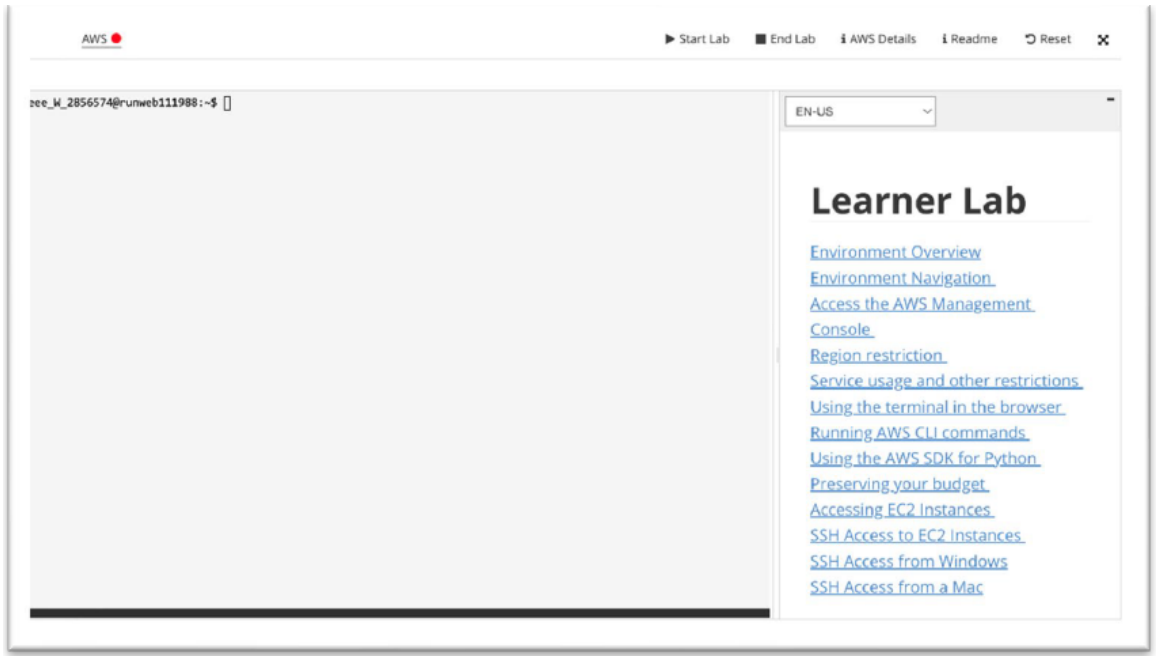
Troubleshooting

1. If you get an error saying your credentials are invalid, when logging in



   Click the logout link and then try again. If that still does not work, try clearing your cache and cookies in your browser. You can also use Incognito mode in your browser or a completely different browser if none of those options work.

2. Take time to read the Learner Lab documentation on the Lab Screen! It is full of helpful information.


AWS
Start Lab
End Lab
AWS Details
Readme
Reset
ee_W_2856574@runweb111988:~$
EN-US
Learner Lab
Environment Overview
Environment Navigation
Access the AWS Management Console
Region restriction
Service usage and other restrictions
Using the terminal in the browser
Running AWS CLI commands
Using the AWS SDK for Python
Preserving your budget
Accessing EC2 Instances
SSH Access to EC2 Instances
SSH Access from Windows
SSH Access from a Mac

Q4 GCP Setup Guide

Getting Started

NOTE:

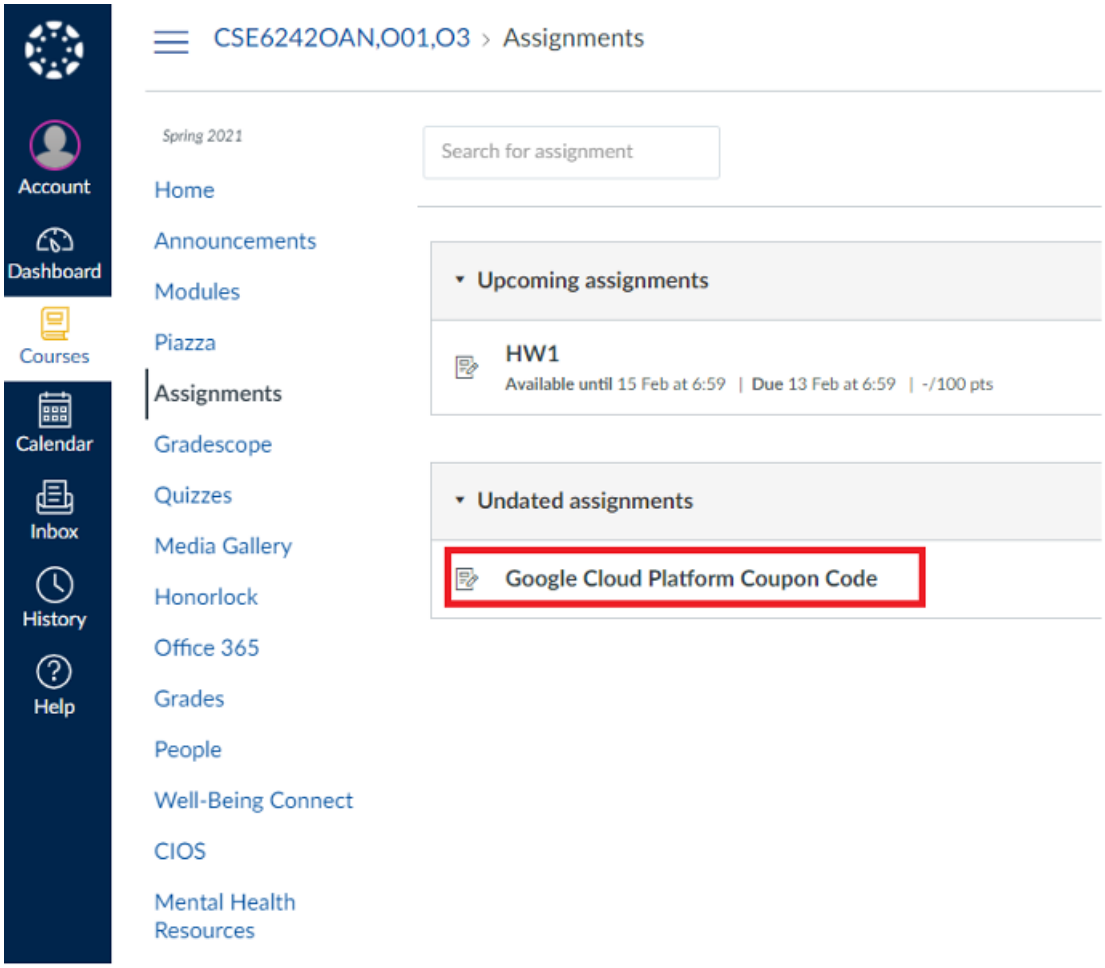
1. Ensure you are using a personal GMAIL account. Do not use GMAIL account associated with any organization or university.
2. This setup guide is designed to help students quickly start GCP Setup and assumes the GMAIL account used is setting up GCP account for the first time. The screens might vary based on your GCP account and this is excepted. Use this guide as a starting point and as a heuristic to identify and setup the GCP Storage buckets and Dataproc Cluster on your own based on the vagaries of your Google Account.
3. To avoid exhausting GCP credits, it is advised to download all notebooks and delete the clusters. Ensure to download all the notebooks from GCP cluster before deleting the GCP cluster (otherwise the work will be lost)

(Students are expected to setup the storage buckets and the clusters on their own. This guide is for reference only, and is not meant to serve as a step-by-step guide due to the variations in individual google accounts)

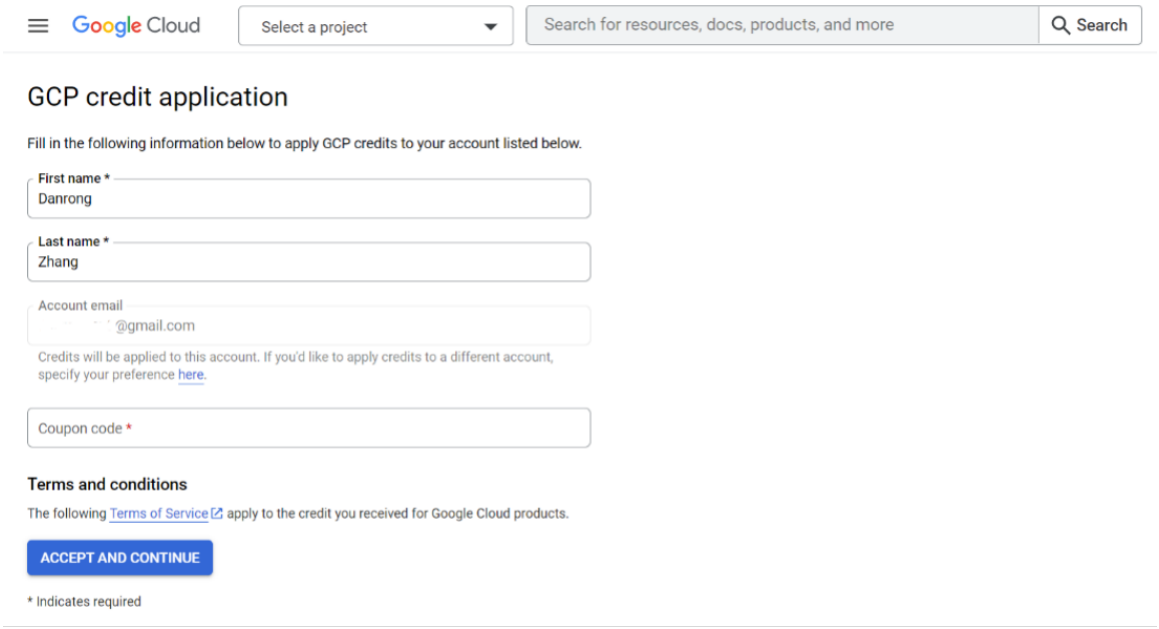
Instructions to setup GCP Storage buckets and Dataproc Cluster are provided below:

Please make sure you have a Google account and if you don't have one, you would need to create a new account.

1. Login into your account using this link
2. Access your GCP coupon through the canvas assignment as shown below.

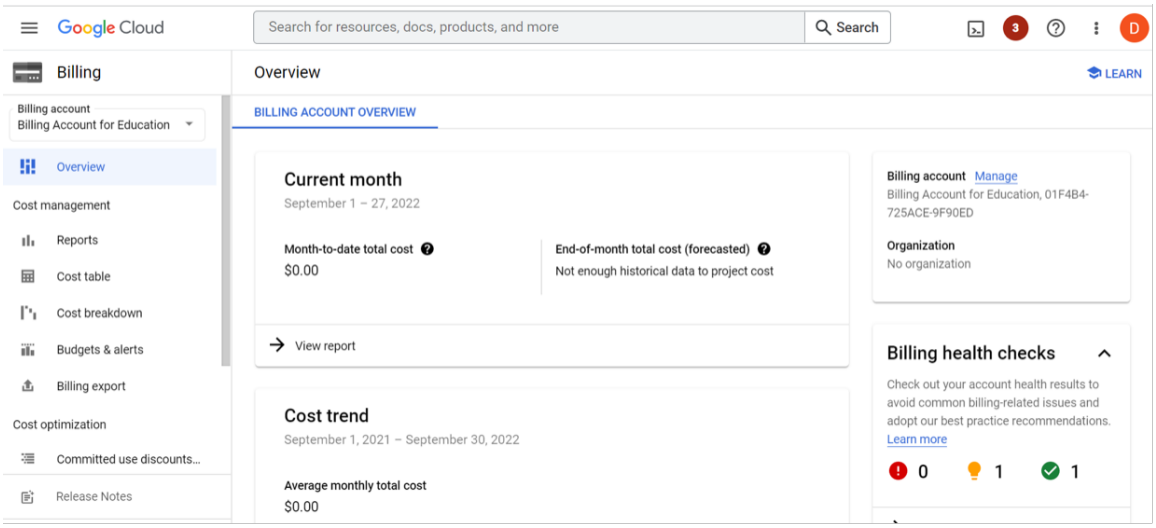


3. Enter your free educational credits using this link.

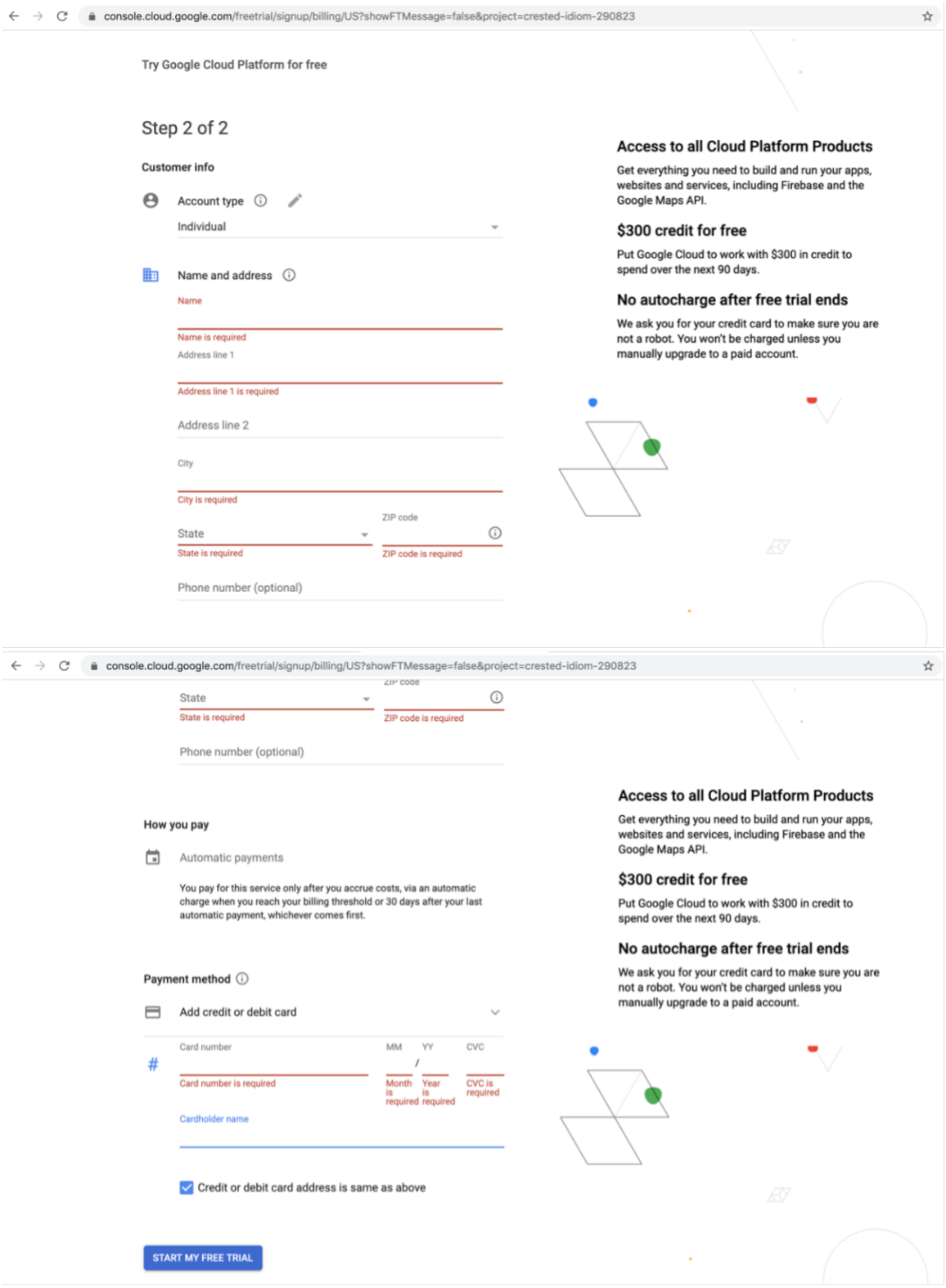


4. When you enter the educational credits, Google automatically creates a new billing account named "Billing Account for Education" and activates the "free trial option" at the top right corner of the console.

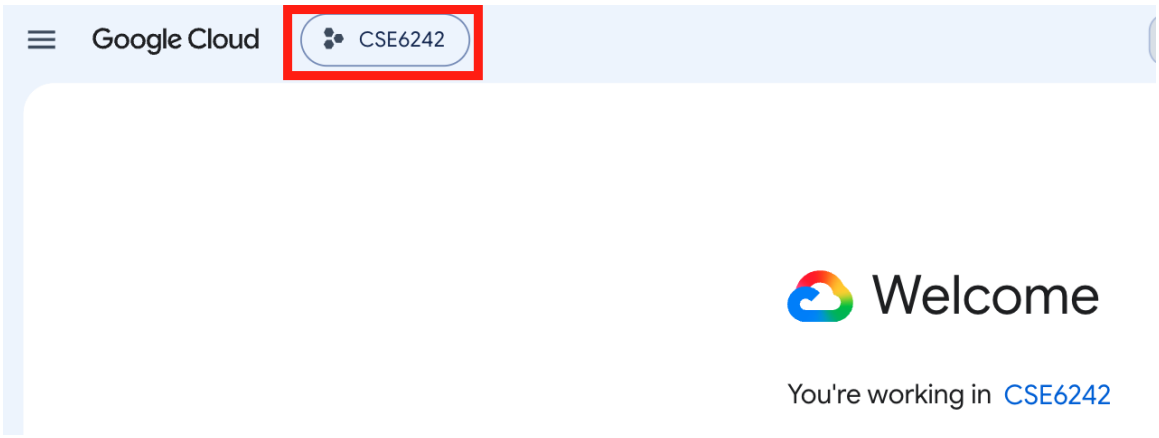
   Note: You might not need to activate if you have previously used GCP with educational credits.



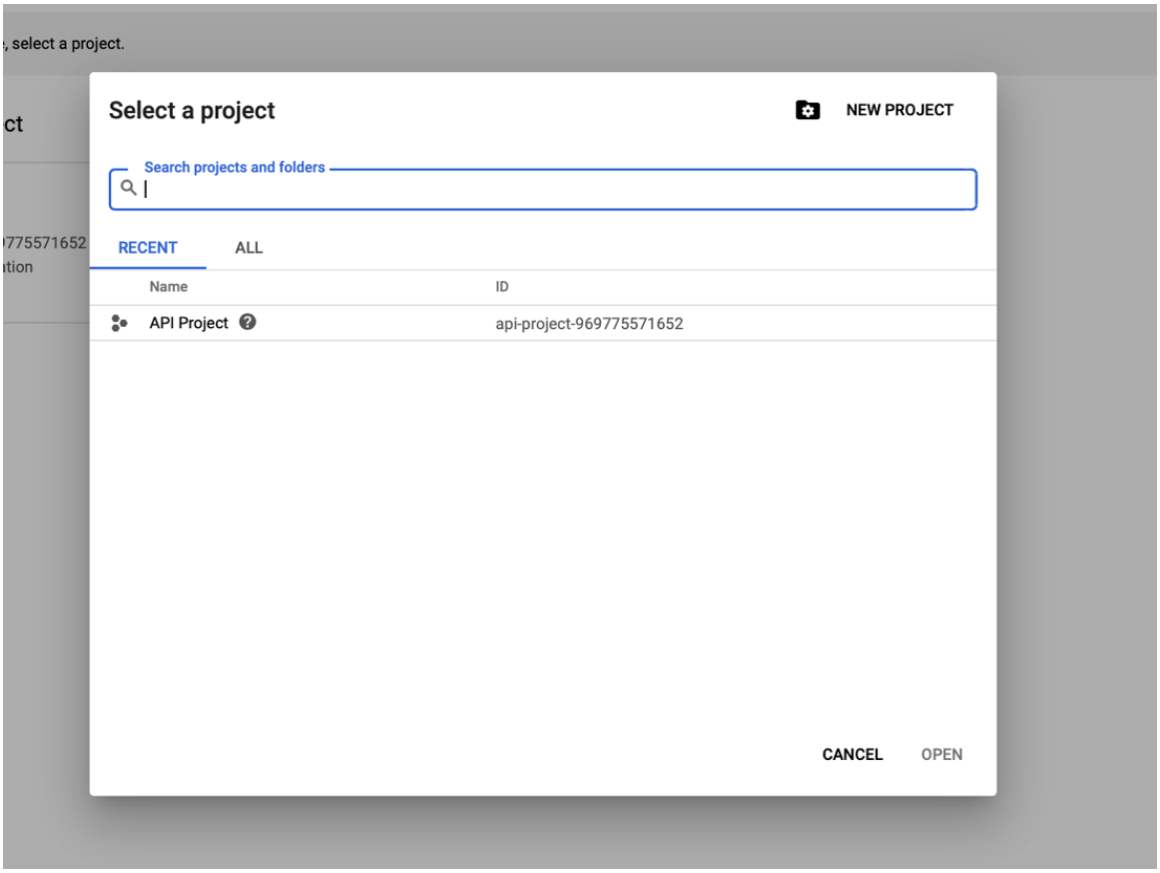
5. You will need to enter default billing credit card information if you are using GCP for the first time. This is to confirm it is a human and you may need to give your personal credit card details for billing (money would not be deducted from your card). Please make sure to use the billing account named "Billing Account for Education" is linked to a new project which will be used to create your storage and clusters (details in step 7). Don't use your own personal billing account otherwise the free educational credits will not be applied.



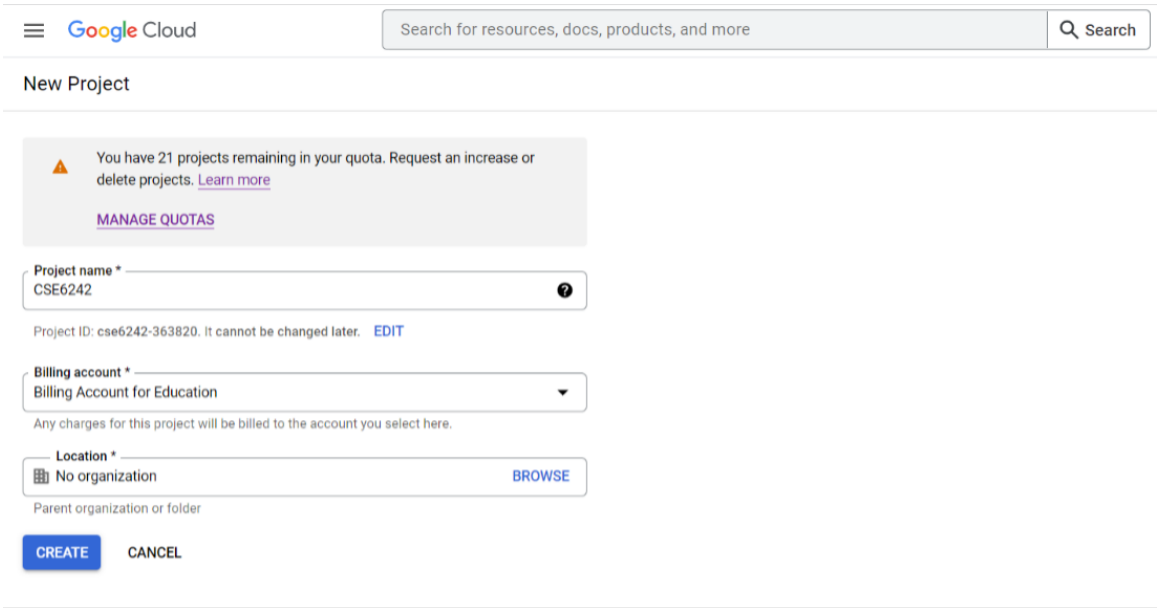
6. Go to Google Console Home
7. On the top left menu, click on button to **"select a project" (or if you already have existing projects, it will list the latest project. Click on the button in this case as well)**



8. Click on New Project on the top right corner

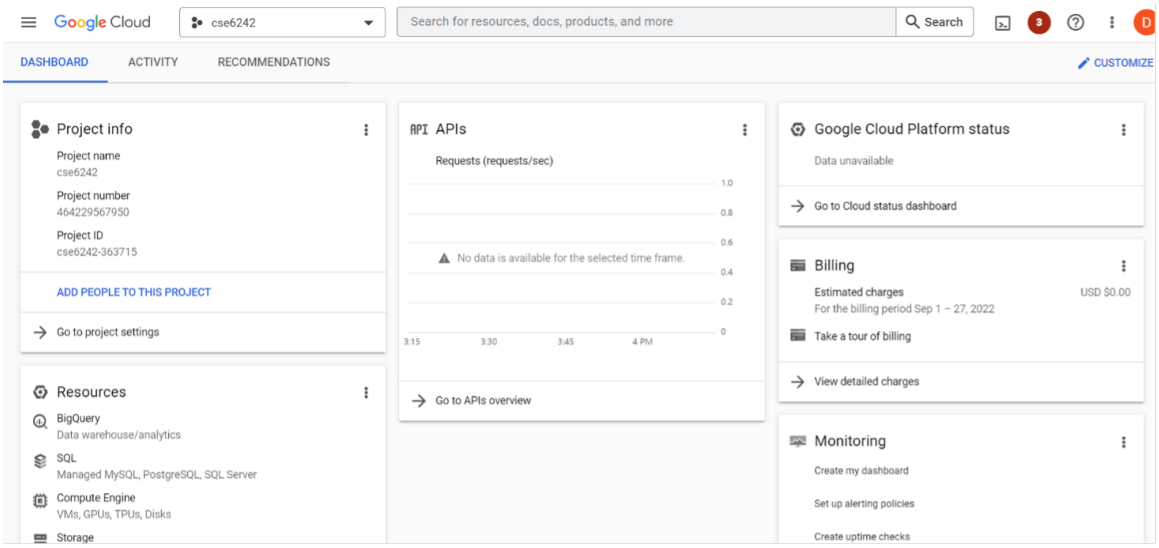


9. Enter CSE6242 (or any other default name) and create a new project.



Note: The new Project CSE6242 should be linked to the Billing Account "Billing Account for Education". This step can be achieved while creating a new project in step 7 as listed above. If you have only one billing account (namely Billing Account "Billing Account for Education", you will not see the 'Billing Account' section and the system will automatically link your project to Billing Account "Billing Account for Education".

10. Wait till the new project is created.

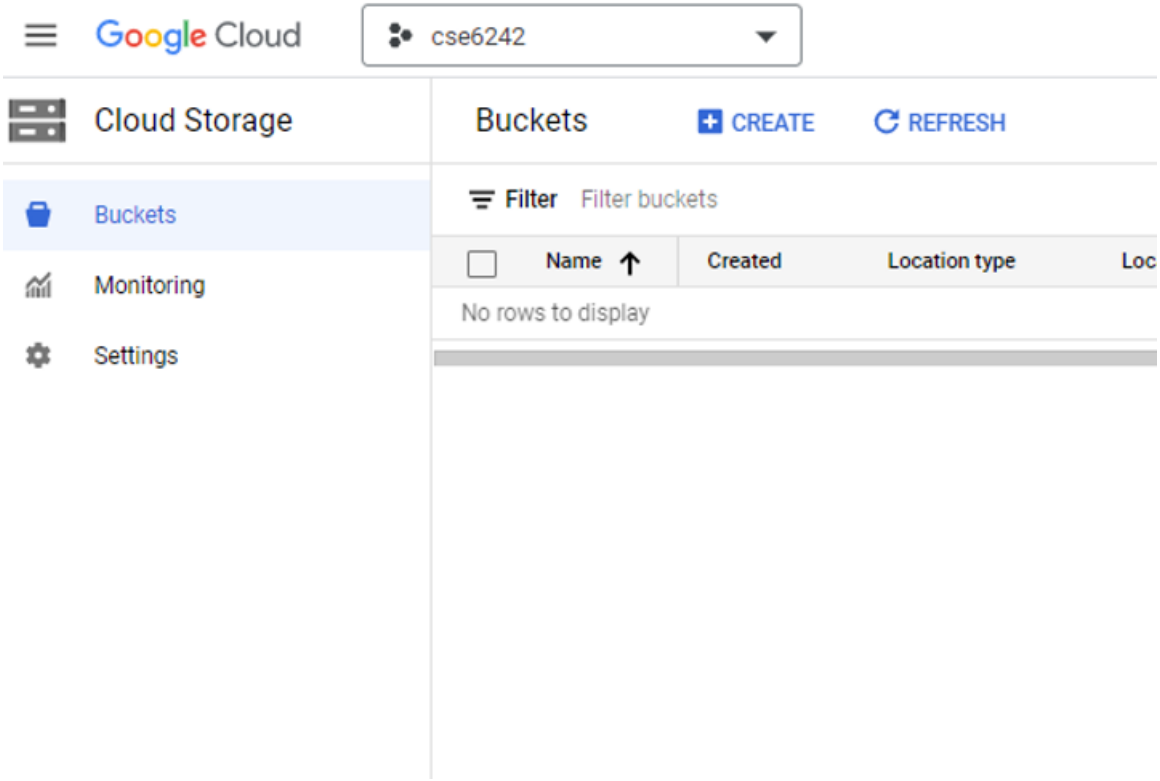


Uploading data files to Spark compatible Google Cloud storage

We have listed the main steps from the documentation for uploading data file to your Google Cloud Platform:

IMPORTANT: The data for this item can be found in this link. You need to download the dataset to your local environment before completing the steps below. They will require you to upload the data linked above from your local environment to GCP storage.

1. Go to Google Storage Home Page
2. On the top left corner, make sure the project is CSE6242 (or whatever you named the project in the previous steps) and then choose '**Cloud Storage**' from the Navigation Menu on the left-hand side.



3. Click **"CREATE"** to create a bucket.
    a) Use your GT Username as bucket name.
    b) Choose Location Type as "Region - Lowest Latency within a Single Region" (to minimize cost) and select the nearest region to you.
    c) Choose "Standard" storage class.
    d) Choose "Access Control" as "Fine Grained"
    e) Other settings - leave them as default.

Once the bucket is created, click "UPLOAD FILES" and select **yellow_tripdata09-08-2021.csv** from your local computer to upload to your bucket. The uploading process may take some time. You will see the upload complete message and you will be able to view the **yellow_tripdata09-08-2021.csv** file in the bucket.

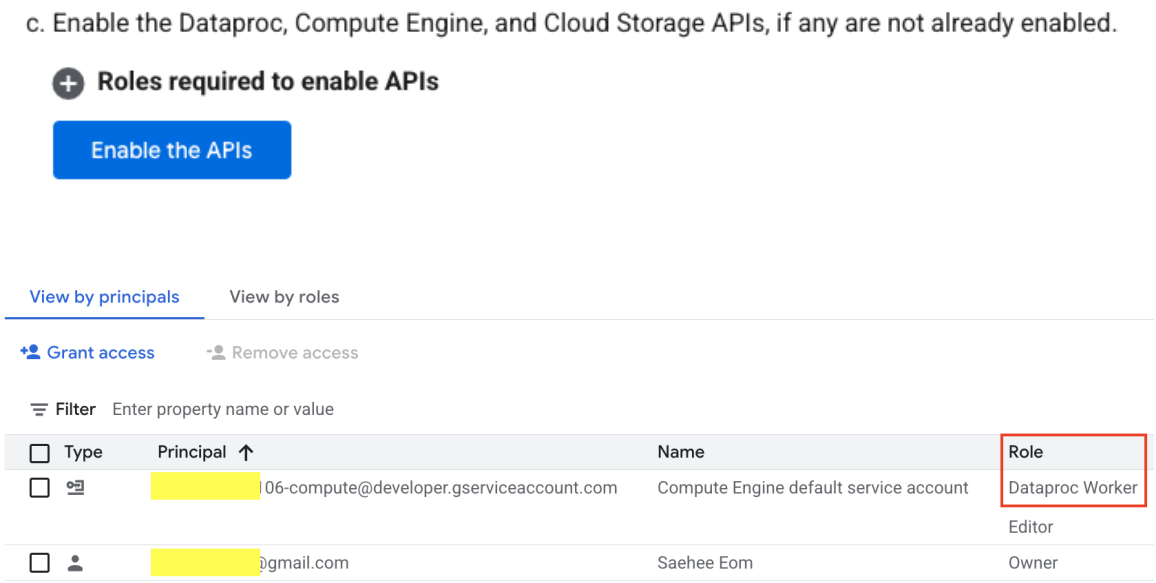
Creating Clusters in Google DataProc (With Spark and Jupyter Notebook Components)

Google DataProc can be used to create an Apache Spark distribution cluster. This means that it handles large amounts of data on demand. The next step is to use Google's DataProc web-based management tool to create a Linux cluster. Follow the recommended steps shown to create a new cluster (or see full documentation here on GCP site).

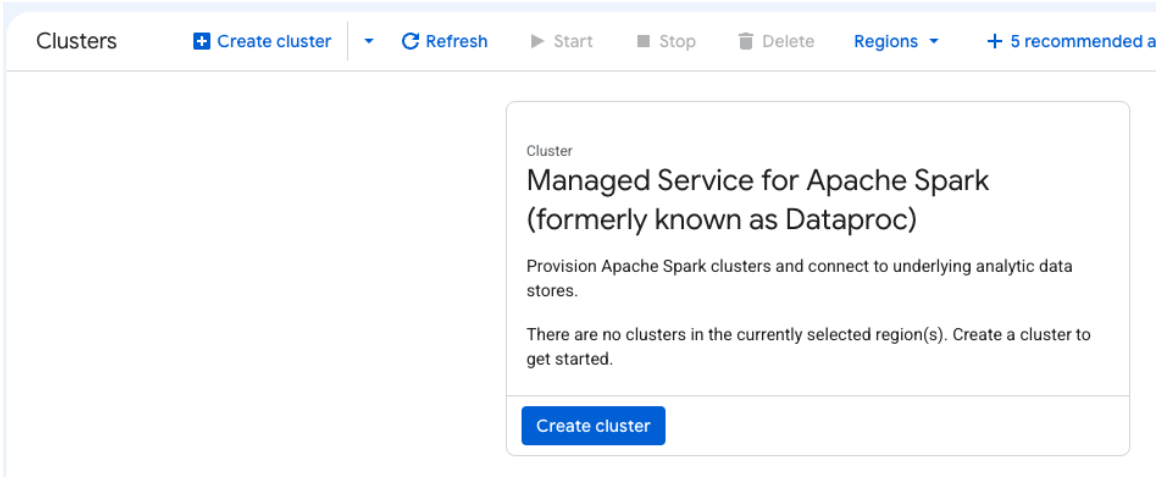
Follow the steps in the below link to create the cluster:

1. Go to Google Compute Engine home page and enable the API (Note that you might not need to enable API if previously used before). This step is to click "Enable the APIs" under step 1c shown in here and following the instruction. NOTE: You may need to add IAM roles to enable APIs. Both Dataproc Worker and Service Usage Admin role is needed for later steps. To do this, enable Cloud Resource Manager API at this link, click on Navigation menu (three line icon on top left), click on 'IAM & Admin' > 'IAM'. Under 'View by principals' click on the pencil icon next to the Service account (check 'Type' field – this account should be labeled as 'Service account' while

your personal gmail account is 'User'). Click '+ Add another role', search 'Dataproc Worker' and Save.

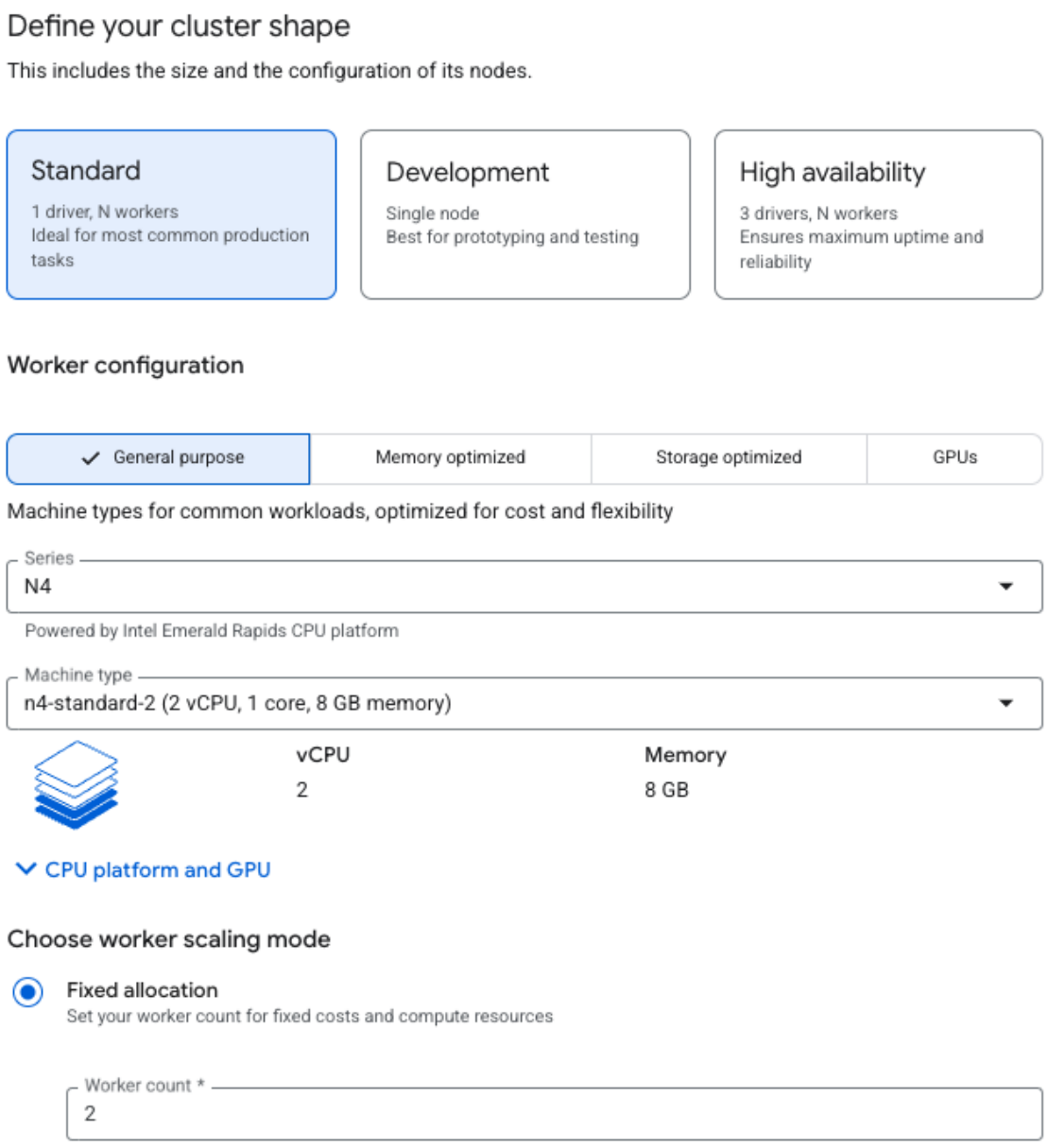


2. Go to Managed Apache Spark (formerly Dataproc) home page
3. On the top left corner, make sure the project is CSE6242 (or whatever you named the project in the previous steps).



4. Enable API and Click 'Create Cluster'. (Note that you might not need to enable API if you previously used Cloud Dataproc before)

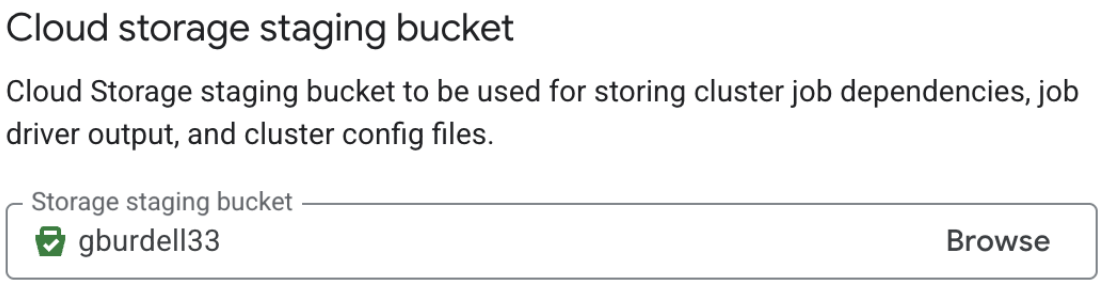
5. Choose your GT Username as the cluster name (or you can use the default name)
6. Select the nearest region to you
7. Choose Cluster type as 'Standard' (1 driver, N workers)
8. Under 'Worker configuration', choose 'General purpose'. Set Series as 'N4' and choose 'n4-standard-2' for the Machine type. (2 vCPU, 1 core, 8 GB memory). Scroll down to 'Additional configuration' toggle. Click the pencil icon next to **'Primary workers'** and set the 'Primary disk size' to **200GB**, then click 'Save'. Next, click the pencil icon next to **'Driver node'**, uncheck **'Default driver node to the same as primary worker'**, set the 'Primary disk size' to **100GB**. (Total must be below 500GB)
9. Choose 'Fixed Allocation' for worker scaling mode and leave 'Worker count' as 2.



10. Under 'Summary', check the Component gateway is 'Enabled' (Enable access to the web interfaces of default and selected optional components on the cluster.)
11. Click the 'Additional configuration' toggle, go to 'Optional Components' section and check 'Jupyter Notebook' is under 'Enabled components'.



12. Click the pencil icon next to 'Customization & Other', set "Cloud Storage staging bucket" field to select your bucket- Browse the name of the bucket you created in prior steps (only specify the name of the bucket which would be your GT Username). Your notebooks will be stored in Cloud Storage under gs://bucket-name/notebooks/jupyter



13. Anything else not mentioned should remain unchanged. Click Create

It will take a few minutes for your Cluster to be up and running. Once the cluster is up and running (you may ignore the message "Sorry, the server was not able to fulfill your request." as long as the status of the cluster is shown as running):

1. Click on your **Cluster Name** to navigate into the cluster details screen
2. Click the **Web Interfaces** tab to display a list of Component Gateway links to the web interfaces of default and optional components installed on the cluster
3. Click the **Jupyter** link. The Jupyter notebook web UI opens in your local browser.
4. **Upload HW3 Q4 skeleton notebook** into your Jupyter notebook web UI (under GCS folder and not under local folder)
5. Open the notebook, set the kernel to **PySpark** and you will be ready to start working on the Q4 section.

Delete Storage and Clusters

After you complete the Q4 section of this homework, please remember to delete both storage and clusters. The deletion process is easy.

NOTE: Ensure to download all the notebooks from GCP cluster before deleting the GCP cluster (otherwise the work will be lost).

1. On the Storage home page, check the bucket you want to delete and then click "DELETE."
2. Wait till the bucket is deleted without any errors.
3. On the Cluster page, check the cluster you want to delete and then click "DELETE."
4. Wait till the cluster is deleted without any errors.